Make one voice easier to follow

UI/UX delivery plan | 3 designers + 4 developers | 2 build days Track: Health & Well-being

The product we are shipping

FOCUS is an installable web app that lets a user upload a short video, select a visible speaker, and listen to the isolated voice. The result includes an original/isolated comparison, a transcript when available, and download and delete controls.

The promise

Help people focus on a chosen speaker in overlapping conversation. Present FOCUS as an assistive listening prototype. Do not claim clinical effectiveness, guaranteed noise removal, or hearing-aid replacement.

The delivery rule

Day 1: a complete, working upload experience. **Day 2:** usability, quality and speed improvements, followed by a gated experiment with processing during recording. The upload experience stays available throughout.

Must ship	Only after the core works
Upload, face selection, processing states, audio comparison, transcript states, download and delete.	Experimental recording, richer motion, word highlighting, extra themes and video export.
Responsive interface, accessible controls, PWA install guidance, refresh recovery and offline messaging.	No accounts, payments, chat, social features or complex onboarding in this build.

Working assumptions

Initial upload limits: 15 seconds and 50 MB, subject to the backend contract. Processing runs on the server/GPU and needs an internet connection. Installing the PWA does not make AI processing work offline.

All H markers below are cumulative working hours across two days: H0-H12 on Day 1 and H12-H24 on Day 2. They are team milestones, not a request to work without breaks.

One journey. Clear states.

Design the full experience, including the moments when the model cannot deliver.

Step	Required design and behavior
1 / Start & upload	A short promise, one primary upload action and a clearly labeled sample option. Show duration/size limits before selection. Provide file preview, replace and remove.
2 / Choose a speaker	Use actual detected face thumbnails and a timestamped preview. Persist a visible selected state. Enable Continue only after selection. Never invent identity labels or imply that arbitrary faces can be selected.
3 / Process	Show real stages: checking file, preparing video, tracking speaker, isolating voice and preparing transcript. Use indeterminate progress unless the backend provides a real percentage.
4 / Listen & compare	Original / Isolated toggle, one shared playback position, play/pause, seek and elapsed time. Never play both tracks together. Keep the selected face visible for context.
5 / Read & export	Audio can become available before the transcript. Show transcript loading, ready or unavailable separately. Offer audio download, transcript export only if supported, and clear delete controls.

Required exceptions

• **Upload:** unsupported format, oversized/overlong file, no audio, corrupt file and interrupted connection. Preserve a useful next action: replace, trim externally or retry.

• **Selection:** no face, multiple faces, unclear thumbnail or lost track. Offer another supported frame only if the backend can use it; otherwise explain the limitation and request another clip.

• **Processing:** waiting in queue, service unavailable, failed job, refresh recovery and offline. Avoid invented ETA values or false completion states.

• **Results:** weak separation, unavailable transcript and deletion pending. Do not label an output as perfect. Let the user compare and try another clip.

Three owners, no collisions

Assign names at kickoff. Each designer has a distinct source of truth and a paired developer.

Owner	Scope and developer pairing
Designer 1 / UX lead	Own journey, flow, wireframes, state matrix, product copy and usability findings. Pair with Dev 2 for API states and Dev 3 for upload/selection.
Designer 2 / Visual system	Own tokens, components, responsive rules and upload/selection/processing high-fidelity screens. Pair with Dev 3.
Designer 3 / Results & QA	Own comparison player, transcript, result actions, install/offline screens, end-to-end prototype and visual QA. Pair with Dev 4; consult Dev 1 on model limits.

Designer 1 / Deliverables

• **By H1:** scope and journey map; confirm model-supported selection with Dev 1. **By H2:** mobile wireframes and complete state/copy matrix. **By H3:** first flow walkthrough with developers.

• **H3-H12:** answer behavior questions, test the working flow and triage blockers. **Day 2:** run short usability sessions, document findings and own the experimental recording flow.

Designer 2 / Deliverables

• **By H2:** typography, color, spacing and core components. **By H4:** upload, selection and processing screens at mobile and desktop sizes. Hand off each screen as soon as it is ready.

• **H4-H12:** component states and implementation checks. **Day 2:** responsive and accessibility fixes; avoid a visual redesign after the core is implemented.

Designer 3 / Deliverables

• **By H3:** player and transcript interaction wireframes. **By H5:** results screens and a clickable end-to-end prototype, including errors. **By H8:** comparison, download and delete behavior approved with Dev 4.

• **H8-H12:** review the live build on a phone. **Day 2:** retest fixes, prepare a truthful demo walkthrough and perform final visual QA.

Design for implementation

Use one shared Figma file. Developers must be able to build before every screen is polished.

File structure and ownership

00 Brief & Flow - Designer 1
01 Tokens & Components - Designer 2
02 Upload / Select / Process - Designer 2
03 Results / PWA States - Designer 3
04 Prototype & QA - Designer 3, with findings from Designer 1

Minimum design system

• One light theme with dark readable text, restrained teal emphasis and clear status colors. Start with one font family and a small type scale. Use an 8-point spacing rhythm and reusable layout rules.

• Components: buttons, file picker, face card, status banner, stage list, player, segmented toggle, transcript panel, toast and confirmation dialog. Specify default, hover, focus, disabled, selected, loading and error states where relevant.

• Design reference widths: 390 px mobile and 1440 px desktop. Annotate fluid behavior and verify the build at 320 px. Use practical touch targets of at least 44 x 44 px for primary controls.

Accessibility acceptance

• Use at least 4.5:1 contrast for normal text. Never communicate state by color alone. Show keyboard focus, label icon buttons and keep the complete flow usable with a keyboard.

• No autoplay. Provide visual feedback for all audio-related actions, readable transcripts when available and reduced-motion behavior. Avoid rapid flashing, ornamental waveforms that imply real analysis, and sound-only alerts.

A screen is ready for development when it includes

1. Named component instances and Auto Layout.
2. Mobile layout plus desktop behavior.
3. Exact labels, limits, error copy and empty/loading states.
4. Interaction notes and the backend state that triggers each variant.
5. Exportable SVG icons/assets, with usage rights checked.
6. A named developer owner and a link to the corresponding task.

Use a Ready for Dev label. Designers do not edit another owner's components without agreement. Record post-handoff changes in a short change log and notify the affected developer.

Ship on Day 1

Rolling handoff and a working build are more valuable than a finished design file waiting for implementation.

Window	Design checkpoint	Build dependency
H0-H1	Confirm scope, owners, user journey and actual model limitations.	Agree upload limits, job stages and speaker selection behavior.
H1-H3	Wireframes, tokens, core components, result interactions and copy.	Developers start from shared components and agreed mock states.
H3-H6	Complete upload/select/process and results handoff; clickable prototype.	A new browser upload must produce real isolated audio by H6.
H6-H9	Review face selection, comparison, transcript states, downloads and deletion.	Fix wrong-speaker selection and broken audio flow before extra styling.
H9-H12	Mobile, keyboard, errors, refresh recovery and PWA guidance reviewed.	External HTTPS link works on another device; freeze the upload baseline.
H12-H15	Observe users, rank friction and verify targeted fixes.	Improve existing behavior; avoid broad redesign.
H15-H18	Design recording experiment only if the core is stable.	ML/backend owners measure latency and quality. Go/no-go at H18.
H18-H20	Finish approved fixes and honest demo screens.	Freeze features at H20.
H20-H24	Final regression review, demo rehearsal and handoff archive.	No new features; keep a working upload fallback.

Coordination rhythm

Use a 10-minute kickoff and a brief checkpoint every three working hours. Each owner reports: what is ready, what blocks a developer, and the next deliverable. Review the live build at H6, H9 and H12; a Figma-only approval is not enough.

If the schedule slips

Cut decorative motion, extra themes, word highlighting, video export and recording first. Preserve upload, real speaker selection, clear wait/error states, comparison, download and delete. If the H6 core path fails, all design effort shifts to unblocking it.

Test, then demonstrate

Success means a new user can complete the task and understand both the result and its limitations.

Quick usability test / Day 2

Designer 1 facilitates three short sessions with people outside the build team where possible. Designer 3 records observations; Designer 2 turns the highest-impact findings into small fixes. This is formative usability feedback, not clinical validation.

• Task: upload a provided clip, choose the requested face, wait for the result, compare both tracks, locate the transcript state, download audio and delete the job. Do not coach unless the participant is stuck.

• Record completion, wrong selections, requests for help, misunderstood waiting states and whether Original vs Isolated is clear. Log issue, severity, owner and retest result. Prioritize anything that blocks completion or misleads the user.

Experimental recording / Gate at H18

Use the label **Experimental: process while recording** until performance justifies a stronger claim. Show camera/microphone permissions, active recording, stop, buffering, lost face and reconnect states. Show measured delay when available. Stopping must release camera and microphone access.

Engineering decides availability using a sustained test, stable queue, measured end-to-end latency and preserved separation quality. The current engineering targets are RTF <= 0.8 and capture-to-play p95 <= 3 seconds; these are prototype targets, not a medical suitability claim. If the gate fails, keep the experiment internal and demonstrate upload.

Demo story / 90 seconds

0-15s: explain the difficulty of following one speaker.
15-35s: upload a short clip and select the visible target.
35-65s: compare original and isolated audio at the same position.
65-80s: show transcript status, download and deletion.
80-90s: state what works today and the next measured improvement.

If a prepared result is used to fit the demo time, label it clearly. Never present a prerecorded output as a fresh live inference. Keep one verified sample as a fallback alongside the real upload path.

Final handoff checklist

Shared Figma link and ownership map; final component library; annotated screens and states; working prototype; SVG assets; resolved/remaining QA issues; demo script; and explicit approval from the paired developers. The live product must match the approved core flow.